

Extension of the PDDG/PM3 and PDDG/MNDO Semiempirical Molecular Orbital Methods to the Halogens

IVAN TUBERT-BROHMAN, CRISTIANO RUCH WERNECK GUIMARÃES,
MATTHEW P. REPASKY,* WILLIAM L. JORGENSEN

Department of Chemistry, Yale University, 225 Prospect St., New Haven Connecticut 06520

Received 13 May 2003; Accepted 3 July 2003

Abstract: The new semiempirical methods, PDDG/PM3 and PDDG/MNDO, have been parameterized for halogens. For comparison, the original MNDO and PM3 were also reoptimized for the halogens using the same training set; these modified methods are referred to as MNDO' and PM3'. For 442 halogen-containing molecules, the smallest mean absolute error (MAE) in heats of formation is obtained with PDDG/PM3 (5.6 kcal/mol), followed by PM3' (6.1 kcal/mol), PDDG/MNDO (6.6 kcal/mol), PM3 (8.1 kcal/mol), MNDO' (8.5 kcal/mol), AM1 (11.1 kcal/mol), and MNDO (14.0 kcal/mol). For normal-valent halogen-containing molecules, the PDDG methods also provide improved heats of formation over MNDO/*d*. Hypervalent compounds were not included in the training set and improvements over the standard NDDO methods with *sp* basis sets were not obtained. For small haloalkanes, the PDDG methods yield more accurate heats of formation than are obtained from density functional theory (DFT) with the B3LYP and B3PW91 functionals using large basis sets. PDDG/PM3 and PM3' also give improved binding energies over the standard NDDO methods for complexes involving halide anions, and they are competitive with B3LYP/6-311++G(d,p) results including thermal corrections. Among the semiempirical methods studied, PDDG/PM3 also generates the best agreement with high-level *ab initio* G2 and CCSD(T) intrinsic activation energies for S_N2 reactions involving methyl halides and halide anions. Finally, the MAEs in ionization potentials, dipole moments, and molecular geometries show that the parameter sets for the PDDG and reoptimized NDDO methods reduce the MAEs in heats of formation without compromising the other important QM observables.

© 2003 Wiley Periodicals, Inc. J Comput Chem 25: 138–150, 2004

Key words: semiempirical MO methods; heat of formation; halogen; PDDG; PM3; MNDO; AM1; NDDO; DFT

Introduction

Semiempirical molecular orbital methods based on the Neglect of Diatomic Differential Overlap (NDDO)¹ approximation, such as MNDO,² AM1,³ and PM3,^{4,5} continue to find widespread use in a variety of applications, including QM/MM simulations of reactions,^{6–18} where they are usually the QM methods of choice. When compared with DFT or *ab initio* methods, semiempirical methods have speed as their main advantage, but larger and more random errors are their main drawbacks. Fifteen years have passed since the publication of the PM3 method, but so far there are no more recent methods in widespread use that give better results. This is surprising considering the concomitant increase in computing power.

In a recent article,¹⁹ our group introduced new NDDO-based methods that make considerable progress for several key deficiencies, particularly addressing isomerization energies, size-scaling problems, and hydrogen-bond energetics. The resulting method

involves addition of small Pairwise Distance Directed Gaussians (PDDG) in the core repulsion functions and the reoptimization of most NDDO parameters. The PDDG/PM3 and PDDG/MNDO methods were parameterized for C, H, N, and O, and tested for a set of 622 molecules, resulting in a decrease of the mean absolute

Correspondence to: W. L. Jorgensen; e-mail: william.jorgensen@yale.edu

*Current address: Schrödinger Inc., 120 West Forty-Fifth St., New York, NY 10036

Contract/grant sponsor: National Science Foundation; contract/grant number: CHE0130996

Contract/grant sponsor: CNPq/Brazil (Conselho Nacional de Desenvolvimento Científico e Tecnológico) (C.R.W.G.)

This article includes Supplementary Material available from the authors upon request or via the Internet at <ftp://ftp.wiley.com/public/journals/jcc/suppmat/25/138> or <http://www.interscience.wiley.com/jpages/0192-8651/suppmat/25/v25.138.html>

error (MAE) in heats of formation from 4.4 to 3.2 kcal/mol, and from 8.4 to 5.2 kcal/mol, respectively, when compared with the original PM3 and MNDO methods. The PDDG approach stemmed from our prior efforts that showed that the use of bond and group equivalents (BGE) to correct semiempirical molecular energies could lead to dramatic improvements in computed heats of formation for CHNO compounds.²⁰ The PDDG Gaussians are based on atom pairs and provide similar corrections, but their application is unrestricted and requires no postprocessing, just as with the original MNDO, AM1, and PM3 methods. In a subsequent study, we applied the BGE method to halogens, silicon, and sulfur.²¹ The BGE results for halogens, which showed 30 to 50% decreases in MAEs for heats of formation, suggested that significant improvements could again be achieved with the PDDG approach.

In this article, the PDDG methods are extended to include compounds with halogens (F, Cl, Br, I). This is an important step in increasing the scope of the methods, particularly for application to druglike molecules. Moreover, it serves as an additional test of the generality and applicability of the PDDG formalism to elements from various rows of the periodic table and with a wide range of electronegativities.

PDDG Formalism

The rationale behind the PDDG formalism has already been described.¹⁹ The key difference between the PDDG/PM3 and PDDG/MNDO methods and their predecessors is the addition of pairwise distance directed Gaussian terms to the core repulsion function [eq. (1)]:

$$PDDG(A, B) = \frac{1}{n_A + n_B} \left[\sum_{i=1}^2 \sum_{j=1}^2 (n_A P_{A_i} + n_B P_{B_j}) \exp \left(-10 \text{Å}^{-2} (R_{AB} - D_{A_i} - D_{B_j})^2 \right) \right] \quad (1)$$

The equation is summed over all distinct atom pairs, A and B , which are separated by a distance R_{AB} in Å; each element requires four parameters P_{A_1} , P_{A_2} , D_{A_1} , and D_{A_2} . The function is weighted using n_A and n_B , which are the number of valence electrons for atoms A and B , respectively. For each pair of atoms there are four Gaussians, which depend only on atomic parameters. The Gaussians are small compared to those used in the original AM1 and PM3 core repulsion functions, and they are centered at distances that correspond to 1,3 or 1,4 interactions, with the aim of representing functional group information that would not be included in the shorter-range Gaussians found in the AM1 and PM3 methods.

Another important difference between the traditional NDDO methods and the PDDG methods is the way in which the molecular energy obtained from the SCF calculation is converted to a heat of formation at 298 K. In both cases, the difference between the heat of formation of each atom and its electronic energy (eisol) is added to the molecular energy. In the traditional method, aisl is obtained as a derived parameter by calculating the energy of the isolated atom with a restricted single-determinant wave function using the desired semiempirical formalism and parameter set;²² for the

PDDG methods, it is treated as an optimizable parameter, obtained from a linear regression so that the conversion from molecular energy to heat of formation gives as small an error as possible.

Parameterization of PDDG Methods for the Halogens

To extend the PDDG formalism to the halogens, the set of parameters for which optimal values were determined in Dewar's methods (U_{ss} , U_{pp} , β_s , β_p , ζ_s , ζ_p , α) and all PM3 Gaussian pre-exponential and distance terms in addition to parameters introduced by the PDDG function (P_{A_1} , P_{A_2} , D_{A_1} , D_{A_2}) were optimized, retaining the values of all one-center two-electron integrals (G_{ss} , G_{pp} , G_{sp} , G_{p2} , H_{sp}) at their standard AM1/MNDO or PM3 values. In order to determine optimized parameter sets, the error function, S , defined in eq. 2, was minimized:

$$S = \sum_i w_i^2 (q_i^{\text{calc}} - q_i^{\text{exp}})^2 + \sum_j w_j \frac{\delta E}{\delta x_j} + (\chi^2)^3 \quad (2)$$

S was minimized in two consecutive steps by application of two optimization algorithms. Due to the large number of molecules for which calculations had to be performed, initial optimization of parameters was performed through application of a gradient-based optimization algorithm, using single-point geometries with molecules at their original semiempirical geometrical minima. In this step, the reference functions, q_i , consisted of heats of formation, ionization potentials, and dipole moments. Exclusively in this step, gradients of the energy with respect to molecular geometry were included in the error function [eq. (2)]. In the second step, molecular geometries were energy-minimized at each optimization cycle; optimization was carried out by simulated annealing and the q_i consisted of heats of formation, ionization potentials, dipole moments, and geometrical variables such as bond lengths and bond angles.

A total of 437 reference values,^{4,23–28} q_i^{exp} , specifically, heats of formation, ionization potentials, dipole moments, and geometrical variables, were used in the derivation of the optimized parameter set. These are predominantly experimental data, but some results of DFT and high-level *ab initio* calculations were also included, namely, eight G2 heats of formation and 37 MP2 and B3LYP bond lengths. The G2 heats of formation and 12 MP2 bond lengths were obtained from the literature.²⁹ The remaining 25 bond lengths were obtained here from B3LYP and MP2 calculations with the 6-311++G(d,p) basis set using GAUSSIAN98.³⁰ A total of 147 molecules formed the training set, which consisted of two cations, seven anions, four S_N2 transition structures, 16 negatively charged complexes, and 118 neutral, closed-shell molecules. All of the latter molecules were employed in the regression to determine values for aisl parameters. The chi-squared (χ^2) term in eq. (2) gives an estimate of the quality of the through-origin multiple regression fit that generates aisl parameters.³¹ Initial testing showed that in order to reduce the role that heats of formation played in the error function, χ^2 had to be taken to the third power instead of to the fourth power, as used by Repasky and coworkers.¹⁹ Also, the original weighting factors (w_i) for dipole moments and ionization potentials¹⁹ were doubled to yield balanced performance for the halogens. Table 1 lists the weighting factors used in the present work.

Table 1. Weighting Factors Used in Parameterization of the PDDG Extension to the Halogens.

Reference function	Weighting factor
Heats of formation	1 mol/kcal
Ionization potentials	20 eV ⁻¹
Dipole moments	40 debye ⁻¹
Bond lengths ^a	100 Å ⁻¹
Bond angles ^a	2/3 deg. ⁻¹
Dihedral angles ^a	1/3 deg. ⁻¹
Gradients for bond lengths ^b	0.1 mol/kcal Å
Gradients for bond angles ^b	0.1 mol/kcal deg.
Gradients for dihedral angles ^b	0.1 mol/kcal deg.

^aFor parameterization using flexible molecular geometries.^bFor parameterization using fixed molecular geometries.

Initial testing showed that reasonable starting points for the PDDG Gaussian distance terms (D_{A_1} , D_{A_2}) were 1.2 and 1.7 Å for fluorine, 1.7 and 2.2 Å for chlorine, 1.9 and 2.4 Å for bromine, and 2.2 and 2.7 Å for iodine. In order to verify if the PDDG functions should be attractive or repulsive, these distance terms were combined with two possible initial guesses for each pre-exponential term (P_A), +0.01 eV and -0.01 eV. As there are two pre-exponential terms (P_{A_1} , P_{A_2}) per element, 256 different initial guesses were scanned using fixed molecular geometries. Taking

the 256 sets of optimized parameters from this first step, they were then reoptimized to minimize the error function using fully flexible molecular geometries. To evaluate the importance of contributions from the PDDG Gaussians, a full parameter optimization with the modified treatment of the eisol parameter was performed in which no PDDG functions were employed for any atom. The resultant reoptimized MNDO and PM3 parameter sets for the halogens yielded methods designated here as MNDO' and PM3'. In this procedure, the MNDO and PM3 parameters for the C, H, N, and O atoms were not reoptimized; little benefit from their reoptimization was previously found.¹⁹

During the parameterization process, the PM3 Gaussian functions showed a strong tendency to become very narrow. A narrow Gaussian only has significant intensity in a very small distance range, and it may become associated with a specific interatomic distance in a specific compound.^{32,33} To prevent this, the PM3 Gaussian widths were constrained to the range 2.0–6.0 Å, as previously suggested by Stewart.⁵

Results and Discussion

Parameters

The optimized parameters for the PM3', PDDG/PM3, MNDO', and PDDG/MNDO methods are listed in Tables 2–5, together with the original PM3⁴ and MNDO^{24–27} parameters for comparison. In

Table 2. Comparison of the PDDG, Reoptimized, and Standard Parameters for Fluorine.^a

	PM3	PM3'	PDDG/PM3	MNDO	MNDO'	PDDG/MNDO
U_{ss}	-110.435303	-112.286530	-111.400432	-131.071548	-133.099116	-134.220379
U_{pp}	-105.685047	-107.014694	-106.395264	-105.782137	-106.422458	-107.155961
β_s	-48.405939	-56.994626	-50.937301	-48.290466	-68.510286	-67.827612
β_p	-27.744660	-28.360714	-31.636976	-36.508540	-38.000501	-40.924818
ζ_s	4.708555	5.340649	5.538033	2.848487	4.149213	4.328519
ζ_p	2.491178	2.373532	2.538066	2.848487	2.925851	2.905042
α	3.358921	3.262762	3.200571	3.419661	3.386133	3.322382
eisol	-437.517170	-448.203985	-442.457133	-476.683781	-482.385667	-488.703243
DD	0.312530	0.250780	0.246601	0.506717	0.378200	0.361556
QQ	0.491633	0.516001	0.482551	0.429963	0.418594	0.421593
ρ_0^b	1.296126	1.296126	1.296126	0.804078	0.804078	0.804078
ρ_1^b	0.738716	0.641526	0.634633	0.460817	0.393260	0.383553
ρ_2^b	0.816987	0.846700	0.805802	0.483389	0.474581	0.476913
a_1	-0.012166	-0.007522	-0.008079			
b_1	6.023574	5.931388	5.938969			
c_1	1.856859	2.001348	1.863949			
a_2	-0.002850	-0.002913	-0.002659			
b_2	6.003720	5.923909	5.925105			
c_2	2.636160	2.486761	2.388864			
P_{A_1}			-0.012866			-0.011579
P_{A_2}			0.007315			-0.012943
D_{A_1}			1.305681			0.834606
D_{A_2}			1.842572			1.875603

^aUnits are as follows: (eV) U_{ss} , U_{pp} , β_s , β_p , eisol, a_1 , a_2 , P_{A_1} , P_{A_2} ; (au) ζ_s , ζ_p ; (Bohr) DD , QQ , ρ_0 , ρ_1 , ρ_2 ; (Å) c_1 , c_2 , D_{A_1} , D_{A_2} ; (Å⁻¹) α , b_1 , b_2 .

^bFor use in MOPAC 6, $\rho_0 = 0.5/AM$, $\rho_1 = 0.5/AD$, $\rho_2 = 0.5/AQ$.

Table 3. Comparison of the PDDG, Reoptimized, and Standard Parameters for Chlorine.^a

	PM3	PM3'	PDDG/PM3	MNDO	MNDO'	PDDG/MNDO
U_{ss}	-100.626750	-94.409371	-95.094434	-100.227166	-107.201721	-111.133653
U_{pp}	-53.614400	-53.825508	-53.921651	-77.378667	-77.959584	-78.062493
β_s	-27.528560	-28.808723	-26.913129	-14.262320	-15.031360	-15.663317
β_p	-11.593920	-13.793851	-14.991178	-14.262320	-14.482267	-15.399331
ζ_s	2.246210	2.408137	2.548268	3.784645	4.011585	4.212404
ζ_p	2.151010	2.243037	2.284624	2.036263	2.043770	2.037647
α	2.517300	2.499127	2.497617	2.542201	2.683552	2.602846
eisol	-315.194950	-303.839634	-305.715201	-353.117667	-370.575497	-378.909727
DD	0.917586	0.865084	0.827561	0.498687	0.451594	0.411609
QQ	0.777923	0.746007	0.732427	0.821760	0.818742	0.821202
ρ_0^b	0.849590	0.849590	0.849590	0.905190	0.905190	0.905190
ρ_1^b	0.733727	0.712147	0.696164	0.620254	0.585355	0.554248
ρ_2^b	1.372235	2.332318	2.299104	0.825977	0.823958	0.825604
a_1	-0.171590	-0.140543	-0.112222			
b_1	6.000800	5.974522	5.963719			
c_1	1.087500	1.013218	1.027719			
a_2	-0.013460	-0.013673	-0.013061			
b_2	1.966620	1.990763	1.999556			
c_2	2.292890	2.352428	2.286377			
P_{A_1}			-0.016552			-0.017119
P_{A_2}			-0.016646			0.005497
D_{A_1}			1.727690			1.466335
D_{A_2}			1.784655			2.236842

^aSee Table 2 for units.^bFor use in MOPAC 6, $\rho_0 = 0.5/AM$, $\rho_1 = 0.5/AD$, $\rho_2 = 0.5/AQ$.**Table 4.** Comparison of the PDDG, Reoptimized, and Standard Parameters for Bromine.^a

	PM3	PM3'	PDDG/PM3	MNDO	MNDO'	PDDG/MNDO
U_{ss}	-116.619310	-115.903230	-115.841963	-99.986441	-111.976464	-100.637007
U_{pp}	-74.227130	-74.174771	-74.205146	-75.671308	-76.246040	-76.015735
β_s	-31.171340	-28.795040	-21.538044	-8.917107	-7.605384	-7.054170
β_p	-6.814010	-8.001009	-8.524764	-9.943740	-10.452577	-10.221030
ζ_s	5.348460	4.832168	4.345079	3.854302	4.526532	3.999975
ζ_p	2.127590	2.197410	2.190961	2.199209	2.238211	2.245040
α	2.511840	2.461357	2.424673	2.445705	2.427809	2.414265
eisol	-352.539900	-350.816635	-351.013887	-346.681250	-373.513808	-349.564096
DD	0.275903	0.374023	0.473860	0.605107	0.444568	0.574623
QQ	0.997053	0.965373	0.968214	0.964587	0.947775	0.944892
ρ_0^b	0.853330	0.853330	0.853330	0.904802	0.904802	0.904802
ρ_1^b	0.740151	0.899918	1.046430	0.688864	0.576478	0.668867
ρ_2^b	1.307627	1.277968	1.280643	0.896927	0.886734	0.884978
a_1	0.960460	0.962806	0.961362			
b_1	5.976510	5.976578	6.013600			
c_1	2.321650	2.331724	2.340445			
a_2	-0.954920	-0.953075	-0.948834			
b_2	5.944700	5.988559	5.976329			
c_2	2.328140	2.333073	2.348745			
P_{A_1}			-0.013772			-0.017133
P_{A_2}			0.008849			-0.016964
D_{A_1}			1.852030			2.201539
D_{A_2}			2.338958			2.255764

^aSee Table 2 for units.^bFor use in MOPAC 6, $\rho_0 = 0.5/AM$, $\rho_1 = 0.5/AD$, $\rho_2 = 0.5/AQ$.

Table 5. Comparison of the PDDG, Reoptimized, and Standard Parameters for Iodine.^a

	PM3	PM3'	PDDG/PM3	MNDO	MNDO'	PDDG/MNDO
U_{ss}	-96.454040	-94.580289	-97.664174	-100.003054	-111.396674	-106.588422
U_{pp}	-61.091580	-61.000631	-61.167137	-74.611469	-75.413181	-75.282605
β_s	-14.494230	-15.073154	-16.592621	-7.414451	-6.553177	-6.698375
β_p	-5.894700	-6.771548	-6.599816	-6.196781	-5.994751	-5.693814
ζ_s	7.001010	5.622416	5.062801	2.272961	2.931141	2.718404
ζ_p	2.454350	2.390751	2.417757	2.169498	2.452085	2.461813
α	1.990190	1.978630	1.978170	2.207320	2.262224	2.242446
eisol	-288.316090	-284.528205	-291.537869	-340.598360	-366.457895	-356.076398
DD	0.158147	0.298557	0.407261	1.425323	1.129270	1.209529
QQ	1.046730	1.074577	1.062574	1.184171	1.047699	1.043559
ρ_0^b	0.998024	0.998024	0.998024	0.904562	0.904562	0.904562
ρ_1^b	0.299417	0.442454	0.532290	1.088506	0.967522	1.002309
ρ_2^b	0.970293	0.986929	0.979783	1.090422	0.994570	0.992001
a_1	-0.131480	-0.117680	-0.136003			
b_1	5.206420	4.418339	3.852912			
c_1	1.748820	1.774648	1.697455			
a_2	-0.036900	-0.034831	-0.037287			
b_2	6.010120	4.814160	5.229264			
c_2	2.710370	2.697421	2.768669			
P_{A_1}			0.012901			0.009616
P_{A_2}			-0.012825			-0.007505
D_{A_1}			1.994299			2.572332
D_{A_2}			2.263417			2.936456

^aSee Table 2 for units.^bFor use in MOPAC 6, $\rho_0 = 0.5/AM$, $\rho_1 = 0.5/AD$, $\rho_2 = 0.5/AQ$.

contrast to what was observed in the parameterization for C, H, N, and O, parameter variations greater than 5% were common. In general, there is greater similarity between the PM3' and PDDG/PM3 parameters than between either of them and the original PM3 values. This suggests that the original parameter set was far from optimal for our error function, and that a significant part of the reduction in the error function was due to changes in NDDO parameters, and not exclusively due to the additional PDDG Gaussian terms. The changes in the parameters from MNDO to MNDO' and PDDG/MNDO were generally larger, probably because the MNDO method does not include any Gaussians in its core repulsion formula and because the present training set differed substantially from the original one.²⁴⁻²⁷

Improved Accuracy in Heats of Formation

The MAEs for the full data set and a number of subsets are given in Table 6. Overall, the PDDG/PM3 method provides the best results, followed by PM3' and PDDG/MNDO, which are significantly better than MNDO' and the original NDDO methods. As expected, MAEs for the training set are lower than those for the test set for all methods that were optimized in this study. The differences between the results for the training and test sets are smaller for the PDDG methods than for the reoptimized NDDO methods.

A cursory look at the results for the various subsets reveals that PDDG/PM3 and PM3' are the best methods for eight subsets each; PDDG/MNDO is the best for six subsets; MNDO' is best for two

subsets; and PM3 and AM1 are best for only one subset each. MNDO is not the best method for any subset. This again suggests that the additional PDDG terms in PDDG/PM3 did not have a large impact on the overall heats of formation, whereas the addition of the Gaussians in PDDG/MNDO has a much greater influence.

Some comments can be made about specific subsets. For example, it appears that the MNDO and especially PM3 parameters for iodine are suboptimal, as pointed out by Tikhomirov and coworkers.³⁴ AM1 and the reoptimized and PDDG methods give much better results for the CHI, CHONI, and aromatic-I subsets. Another set that shows a wide range of results is the CHF ($N_F \geq 4$) set, which includes several large perfluorinated molecules. AM1 and MNDO yield large systematic errors for these molecules, mostly because of scaling problems discussed in the previous article, which also affected large hydrocarbons.¹⁹ The error decreases from 47.3 kcal/mol for MNDO to 19.0 kcal/mol for MNDO', suggesting that a main problem was the limited size of the molecules used for the original parameterization. The addition of either the PM3 or PDDG Gaussians reduces the error further to 12.8 kcal/mol for PM3, 9.6 kcal/mol for PM3', and 13.5 kcal/mol for PDDG/MNDO. Having both PM3 and PDDG Gaussians does not result in further improvement; the MAE of 11.7 kcal/mol for PDDG/PM3 is between those for PM3 and PM3'. Note that the OCX subset (halogen alpha to C—O single bond) includes mostly perfluorinated ethers, so the same argument applies here too. The poorer performance for the polyfluoro compounds is the key

Table 6. Mean Absolute Errors in Calculated Heats of Formation (kcal/mol).

Class	N^a	PDDG		Reoptimized NDDO		Standard NDDO		
		PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO
Overall	442	5.6	6.6	6.1	8.5	8.1	11.1	14.0
Training set	147	5.3	6.4	5.7	6.9	9.1	11.5	12.9
Test set	295	5.7	6.7	6.4	9.3	7.6	11.0	14.6
CHF ($N_F \leq 3$)	31	4.9	5.7	4.1	4.6	6.5	7.4	6.7
CHF ($N_F \geq 4$)	27	11.7	13.5	9.6	19.0	12.8	27.5	47.3
CHCl	64	4.0	5.3	5.6	5.2	6.0	6.0	6.7
CHBr	40	3.3	4.0	4.4	4.7	5.5	5.4	4.4
CHI	30	2.9	2.7	3.1	3.2	8.1	3.8	7.0
CHONF	45	7.5	9.5	8.6	19.3	7.4	15.1	27.4
CHONCl	47	4.5	5.0	5.3	6.4	6.6	6.4	5.9
CHONBr	11	4.3	3.4	4.1	4.6	5.7	6.5	6.0
CHONI	10	5.2	4.9	5.4	5.7	8.3	6.0	7.9
CHXY	44	3.7	5.0	5.8	5.6	5.5	9.7	11.7
CHONXY	9	8.0	7.9	7.2	6.5	13.6	23.5	38.1
Inorganic	29	6.7	8.1	7.4	11.4	9.4	7.4	11.5
Aromatic	77	6.0	6.3	6.4	6.5	7.6	6.3	7.6
Alpha carbonyl	14	4.8	5.0	5.4	6.3	6.7	8.0	11.7
OCX	13	10.1	9.1	9.1	9.9	12.5	23.3	45.4
Allylic	11	7.3	9.1	5.7	11.9	9.0	18.1	33.0
Benzylic	11	9.2	9.3	10.4	11.1	12.8	14.4	18.1
Aromatic-F	20	11.1	14.2	11.5	14.1	12.2	11.0	15.4
Aromatic-Cl	37	4.2	3.7	5.0	3.6	4.8	4.9	4.0
Aromatic-Br	11	5.2	5.0	5.3	5.9	7.4	5.9	6.7
Aromatic-I	12	6.0	6.1	5.8	7.0	9.7	6.0	11.4
Anions	15	9.6	14.2	9.2	12.4	9.6	16.6	15.5
Complexes	23	6.2	7.2	5.9	8.9	15.4	26.1	22.7
Transition states	6	12.7	3.9	14.1	4.6	26.6	44.1	33.3
Cations	11	7.6	11.3	6.9	12.5	8.1	14.3	8.2

^a N is the number of molecules in the class.

reason for the final MAEs being higher than for the CHNO compounds.¹⁹ The combination of PDDG and PM3 Gaussians, however, is beneficial for the CHXY subset, that is, halogenated hydrocarbons with more than one kind of halogen. While PM3' has a MAE of 5.8 kcal/mol and PDDG/MNDO yields 5.0 kcal/mol, PDDG/PM3 has a MAE of only 3.7 kcal/mol. A similar argument may be made about the CHCl and CHBr subsets.

For the set of 11 cations, MAEs of 7.6 and 11.3 kcal/mol are found for PDDG/PM3 and PDDG/MNDO. MNDO, AM1, and PM3 give MAEs of 8.2, 14.3, and 8.1 kcal/mol, respectively. PDDG/PM3 generates MAEs that are 6% smaller than PM3, while PDDG/MNDO generates MAEs that are 21% smaller than AM1 and 38% larger than MNDO. In comparing the PDDG and the reoptimized NDDO methods, PDDG/PM3 generates errors that are 10% larger than PM3', while PDDG/MNDO gives a small improvement of 10% over MNDO'. The fact that the PDDG/MNDO and MNDO' are outperformed by MNDO for cations and the small improvement given by PDDG/PM3 and PM3' over PM3 for this class may be attributed to the inclusion of only two cations out of 147 molecules in the training set. For the full set of 15 anions, MAEs of 9.6 and 14.2 kcal/mol are obtained for PDDG/PM3 and PDDG/MNDO. MNDO, AM1, and PM3 generate MAEs of 15.5, 16.6, and 9.6 kcal/mol. While PDDG/PM3 gives the same MAE as

PM3, PDDG/MNDO provides MAEs that are 14 and 8% smaller than AM1 and MNDO, respectively. Compared to the reoptimized NDDO methods, PDDG/PM3 generates errors that are 4% larger than PM3' and PDDG/MNDO generates errors that are 15% larger than MNDO'. Although the errors for the set of 15 anions are relatively large using all semiempirical formalisms, Table 7 shows that significantly improved heats of formation are obtained for the halide anions by the PDDG and the reoptimized NDDO methods.

Comparison with MNDO containing *d* orbitals (MNDO/*d*) was also performed. For the set of 155 normal-valent halogen-containing molecules selected by Thiel and Voityuk in their work,³⁵ MAEs in heats of formation of 3.9, 5.3, 5.3, and 6.3 kcal/mol were obtained for MNDO/*d*, MNDO, AM1, and PM3, respectively. PM3' and MNDO', with MAEs of 4.0 and 4.1 kcal/mol, provide more accurate heats of formation for this specific data set than the semiempirical methods with an *sp* basis set, but are outperformed by MNDO/*d*. However, even employing only an *sp* basis set, PDDG/PM3 and PDDG/MNDO, with MAEs of 3.3 and 3.4 kcal/mol, perform better than MNDO/*d*. The PDDG and reoptimized NDDO methods were also subsequently tested for some hypervalent halogen compounds, for example, ClF₃, BrF₃, ClF₅, BrF₅, IF₅, IF₇, ClF₃O, BrFO₃, and ClFO₃. Overall, the resultant heats of formation and geometries for hypervalent compounds are as poor

Table 7. Experimental and Calculated Heats of Formation for the Halide Anions (kcal/mol).

Anion	PDDG		Reoptimized NDDO		Standard NDDO			Exp. ^a
	PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO	
F [−]	−60.1	−75.3	−54.1	−67.8	−31.2	3.4	−17.1	−61.0
Cl [−]	−57.2	−57.5	−55.5	−54.2	−51.2	−37.7	−54.7	−55.9
Br [−]	−52.5	−48.7	−55.7	−51.2	−56.2	−20.4	−37.5	−52.3
I [−]	−56.6	−46.1	−53.0	−46.5	−64.6	−2.2	−6.5	−46.5

^aRef. 4.

as those provided by the standard NDDO methods and much worse than those from MNDO/*d*.³⁵ As hypervalent compounds were not included in the present training set, it is not possible to state whether an *spd* basis set is essential to treat these molecules. It should also be realized that MNDO/*d* still suffers from the intrinsic problems of MNDO for the first-row part of organic molecules. For example, the branching errors are still significant; the MNDO/*d* errors in heats of formation for 1-chlorobutane, 2-chlorobutane, isobutyl chloride, and *t*-butyl chloride are 1.2, 5.6, 6.4, and 12.5 kcal/mol, while they are 1.9, 0.4, 0.6, and 3.8 kcal/mol with PDDG/PM3.

Comparison with results for heats of formation from DFT methods was facilitated using recent data from Cloud and Schwartz.³⁶ Their results did not become known to us until after the parameterization for this article was completed. The purpose of their DFT study was to evaluate the performance of the B3LYP and B3PW91 density functionals in conjunction with the 6-311+G(3df,2p) and 6-311G(d,p) basis sets for computation of enthalpies of formation for 32 haloalkanes. As summarized in Table 8, the PDDG MAEs in heats of formation for the halomethanes are competitive with those from both DFT methods using the large basis set, and they are considerably better than the MAEs using the 6-311G(d,p) alternative. In going from the halomethane to the haloethane set, errors from B3LYP and B3PW91 with both basis sets significantly expand, reflecting size consistency problems. The PDDG errors, on the other hand, lessen because molecules of various sizes were included in the training set. Overall, PDDG/PM3 and PDDG/MNDO perform better than B3PW91 with the smaller basis set and than B3LYP with both basis sets, and the

PDDG methods are competitive with B3PW91 with the large basis set. Of course, the PDDG calculations enjoy a speed advantage of many orders of magnitude over the DFT alternatives. In comparison with standard NDDO methods, the PDDG MAEs for the halomethane set are similar to those from AM1, MNDO, and PM3. However, for the haloethane set, significant improvement over the traditional semiempirical methods is observed. The identical MAEs for PDDG/PM3 and PM3' also show that most of the improvement with respect to PM3 comes from the reoptimization of the PM3 parameters and modification to the treatment of the eisol parameter. Conversely, most of the improvement observed for PDDG/MNDO when compared to MNDO may be attributed to the PDDG function in view of the larger MAEs observed for MNDO'.

Binding Energies

Ion-molecule interaction energies for 22 negatively charged complexes were computed using all of the semiempirical methods. Experimental and *ab initio* G2 and CCSD(T) binding energies were used as references (Table 9).^{29,37–40} Even with the inclusion of 16 negatively charged complexes in the training set, larger MAEs in binding energies were obtained for PDDG/MNDO and MNDO' than for MNDO. PDDG/MNDO and MNDO' considerably underestimate the attractions, especially for hydrogen-bonded complexes containing fluoride. This derives from both overly negative heats of formation for fluoride, compared to the experimental value (Table 7), and underestimation of the hydrogen-bond strengths for the complexes. Significant improvement over PM3 is

Table 8. Comparison of Mean Absolute Errors in Heats of Formation from DFT and Semiempirical Methods (kcal/mol).

	<i>N</i> ^b	DFT ^a				PDDG		Reoptimized NDDO		Standard NDDO		
		6-311+G(3df,2p)		6-311G(d,p)		PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO
		B3LYP	B3PW91	B3LYP	B3PW91							
Halomethanes	14	4.2	3.1	12.0	5.7	4.2	4.0	4.2	8.7	4.0	3.8	3.8
Haloethanes	18	7.6	3.5	15.9	6.4	2.4	3.5	2.4	6.8	4.3	3.6	5.5
Overall	32	6.1	3.4	14.2	6.1	3.2	3.7	3.2	7.6	4.2	3.7	4.7

^aRef. 36.^b*N* is the number of molecules in the class.

Table 9. Enthalpies of Binding from DFT and Semiempirical Methods Compared to Best-Available Reference Values (kcal/mol).

Complex	DFT	PDDG		Reoptimized NDDO		Standard NDDO			Ref.
	B3LYP ^a	PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO	
C ₆ H ₅ OH...F ^{-b}	-50.3	-41.4	-17.0	-49.6	-21.8	-70.4	-108.0	-65.4	-41.3 ^c
C ₆ H ₅ OH...F ^{-d}	-29.4	-22.7	-6.8	-20.3	-6.8	-16.5	-11.6	-7.3	-19.6 ^c
HCOOH...F ^{-e}	-53.7	-38.0	-7.4	-46.7	-16.8	-67.2	-104.5	-60.5	-45.3 ^c
HCOOH...F ^{-f}	-27.1	-26.0	-6.1	-23.3	-8.3	-19.1	-14.8	-8.9	-19.0 ^c
CH ₃ COOH...F ^{-g}	-49.7	-39.5	-8.6	-47.1	-16.6	-67.7	-104.8	-60.4	-44.1 ^c
CH ₃ COOH...F ^{-h}	-27.8	-24.5	-5.5	-22.6	-8.4	-18.6	-14.6	-8.9	-21.1 ^c
CH ₃ CH ₂ OH...F ⁻	-32.7	-25.8	-9.2	-27.0	-7.6	-43.4	-76.2	-35.0	-31.5 ^c
F ⁻ ...HF	-46.7	-41.1	-19.5	-39.6	-19.0	-37.8	-24.4	-28.5	-38.5 ^c
F ⁻ ...H ₂ O	-29.1	-25.5	-9.5	-26.1	-9.1	-28.7	-53.8	-8.7	-23.3 ^c
F ⁻ ...HCN	-46.7	-47.2	-19.9	-45.9	-11.9	-63.5	-80.2	-39.6	-39.5 ^c
F ⁻ ...CH ₃ F ⁱ	-14.6	-9.0	-9.1	-9.1	-7.0	-8.1	-9.8	-5.8	-13.7 ^j
F ⁻ ...CH ₃ Cl ⁱ	-17.1 ^k	-9.9	-9.9	-11.1	-9.2	-10.1	-14.1	-8.5	-15.9 ^l
Cl ⁻ ...C ₆ H ₆	-7.1	-8.2	-4.4	-14.1	-2.7	-13.1	-10.0	-2.4	-9.9 ^m
Cl ⁻ ...C ₆ H ₅ Cl	-9.7	-10.6	-8.2	-17.7	-6.2	-16.0	-13.4	-5.5	-13.6 ^m
HCOOH...Cl ⁻	-21.9	-23.3	-8.5	-23.7	-8.4	-22.2	-18.9	-7.1	-25.6 ^m
CH ₃ CH ₂ OH...Cl ⁻	-15.3	-17.0	-7.3	-19.0	-7.1	-17.8	-13.9	-6.4	-17.3 ^m
Cl ⁻ ...HCl	-27.5	-42.3	-29.4	-40.4	-29.4	-39.3	-32.1	-24.8	-23.1 ^m
Cl ⁻ ...H ₂ O	-14.5	-17.1	-8.3	-18.3	-8.1	-17.3	-13.8	-7.6	-14.4 ^m
Cl ⁻ ...HCN	-21.3	-37.2	-18.1	-31.6	-12.0	-30.2	-22.4	-11.1	-21.0 ^m
Cl ⁻ ...CH ₃ F ⁱ	-8.6 ^k	-11.6	-7.7	-8.1	-6.4	-6.9	-6.9	-5.1	-9.5 ^l
Cl ⁻ ...CH ₃ Cl ⁱ	-9.8	-12.4	-8.9	-9.8	-8.5	-8.9	-8.6	-7.3	-10.4 ^j
Cl ⁻ ...CH ₃ Br ⁱ	-10.6 ⁿ	-12.8	-8.4	-10.5	-8.7	-12.2	-8.7	-6.2	-11.2 ^o
Cl ⁻ ...CH ₃ CN	-15.7	-23.4	-9.4	-19.6	-6.5	-18.7	-15.0	-6.2	-15.8 ^m
Br ⁻ ...CH ₃ Cl ⁱ	-8.6 ⁿ	-11.6	-10.8	-7.3	-7.6	-10.2	-7.5	-7.1	-10.9 ^o
Br ⁻ ...CH ₃ Br ⁱ	-4.1	-12.2	-10.2	-7.7	-7.8	-14.4	-7.6	-6.0	-9.7 ^j
MAE (kcal/mol)	4.0	4.7	11.6	4.0	11.8	7.4	15.0	9.1	

^aBasis set is 6-311++G(d,p). B3LYP binding energies include thermal corrections and the *PV* term. The zero point energy corrections were scaled by the empirical factor 0.98.

^bResult for C₆H₅OH + F⁻ → C₆H₅OH...F⁻.

^cExperimental results at room temperature. Ref. 37.

^dResult for C₆H₅O⁻ + HF → C₆H₅OH...F⁻.

^eResult for HCOOH + F⁻ → HCOOH...F⁻.

^fResult for HCOO⁻ + HF → HCOOH...F⁻.

^gResult for CH₃COOH + F⁻ → CH₃COOH...F⁻.

^hResult for CH₃COO⁻ + HF → CH₃COOH...F⁻.

ⁱColinear complexes.

^jG2 value at 298 K. Ref. 29.

^kB3LYP binding energy was zero-point corrected using the scaling factor 0.98. No thermal corrections or *PV* term was included to be consistent with the reference value.

^lBasis set is aug-cc-pVTZ. CCSD(T) binding energies were zero-point corrected with MP2 frequencies, but do not include thermal corrections or the *PV* term. Ref. 39.

^mExperimental binding energies at room temperature. Ref. 38.

ⁿB3LYP binding energy was not zero-point corrected. Thermal corrections and the *PV* term were not included to be consistent with the reference value.

^oBasis set of 432 contracted Gaussian orbitals was used. CCSD(T) binding energies do not include zero-point corrections, thermal corrections, or the *PV* term. Ref. 40.

obtained for PDDG/PM3 and PM3' (Table IX). PDDG/PM3 and PM3' generate MAEs in binding energies that are 36 and 46% smaller than PM3. Consistently, it is important for this application that the errors in heats of formation for F⁻, Cl⁻, and Br⁻ in Table 7 are particularly small with PDDG/PM3; the error for I⁻ is significantly larger and the accuracy of absolute binding energies

for iodide complexes may be lessened. Structures for the complexes and the separated species were also energy-minimized in this work using B3LYP with the 6-311++G(d,p) basis set, that is, B3LYP/6-311++G(d,p)//B3LYP/6-311++G(d,p) calculations were performed.³⁰ The resultant B3LYP binding energies in Table 9 also include zero-point and thermal corrections and a ΔPV term

Table 10. Intrinsic Activation Barriers for S_N2 Reactions from NDDO Methods Compared to Reference Values (kcal/mol).

Reaction	PDDG		Reoptimized NDDO		Standard NDDO			Ref.
	PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO	
$F^- \dots CH_3F \rightarrow [F \dots CH_3 \dots F]^-$	43.91	37.52	39.22	34.02	45.17	28.57	44.93	11.02 ^a
$Cl^- \dots CH_3Cl \rightarrow [Cl \dots CH_3 \dots Cl]^-$	15.66	13.88	12.33	7.05	9.64	9.05	10.51	12.79 ^a
$Br^- \dots CH_3Br \rightarrow [Br \dots CH_3 \dots Br]^-$	9.82	11.47	9.84	5.81	2.69	5.50	8.03	10.76 ^a
$I^- \dots CH_3I \rightarrow [I \dots CH_3 \dots I]^-$	13.19	14.40	4.43	6.77	5.29	2.43	10.11	9.75 ^a
$F^- \dots CH_3Cl \rightarrow [F \dots CH_3 \dots Cl]^-$	12.91	15.55	8.20	9.55	5.55	0.07	5.1	2.88 ^b
$Cl^- \dots CH_3F \rightarrow [Cl \dots CH_3 \dots F]^-$	47.58	36.62	45.64	35.41	61.39	76.12	77.68	27.71 ^b
$Br^- \dots CH_3Cl \rightarrow [Br \dots CH_3 \dots Cl]^-$	16.56	16.09	14.30	12.29	16.29	5.39	7.35	16.05 ^c
$Cl^- \dots CH_3Br \rightarrow [Cl \dots CH_3 \dots Br]^-$	10.84	10.59	8.30	2.63	0.55	11.09	11.5	9.10 ^c

^aG2 intrinsic barriers at 298 K. Ref. 29.^bBasis set is aug-cc-pVTZ. CCSD(T) intrinsic barriers at 0 K were zero-point corrected with MP2 frequencies. Ref. 39.^cBasis set of 432 contracted Gaussian orbitals was used. CCSD(T) intrinsic barriers at 0 K do not include zero-point corrections. Ref. 40.

of $-RT$. The zero-point energies were scaled by the empirical factor 0.98, as proposed by Bauschlicher.⁴¹ Alternatively, the zero-point and thermal corrections were made to the B3LYP results in this manner, except vibrational frequencies below 600 cm^{-1} were treated as classical rotations; however, this led to a larger MAE, 4.3 kcal/mol. Overall, the quality of the results from the PDDG/PM3, PM3', and DFT methods is similar; the MAEs are 4.7, 4.0, and 4.0 kcal/mol, respectively.

S_N2 Reactions

Intrinsic activation energies for eight S_N2 reactions involving methyl halides and halide anions were computed using the semiempirical methods, as summarized in Table 10. Results from *ab initio* G2 and CCSD(T) calculations are provided for reference.^{29,39,40} The intrinsic activation energy is the height of the barrier between the ion-dipole complexes; the well-depths for the ion-dipole complexes are in Table 9. It should be noted that only identity S_N2 transition structures involving methyl halides and halide anions were included in the training set. Table 10 shows that the semiempirical methods greatly overestimate the activation barriers for S_N2 reactions in which fluoride is the nucleophile or leaving group. On the other hand, the PDDG methods and PM3' yield substantial improvement over the standard NDDO methods for the S_N2 reactions of methyl chloride and methyl bromide with chloride and bromide ions. MNDO' consistently underestimates the intrinsic barriers for these reactions. For the identity S_N2 reaction involving methyl iodide and iodide anion, the PDDG/PM3 result better agrees with the reference than both PM3 and PM3', which underestimate the barrier. Overall, the results in Table 10 for reactions involving Cl, Br, and I highlight the value of including transition structures in the parameterization of semiempirical methods. These methods are capable of a good representation of the electronics of five-coordinate carbon.

Treatment of Other QM Observables

Ionization potentials, dipole moments, and molecular geometries arising from the standard NDDO methods are generally

acceptable for most problems. Therefore, during the derivation of the parameter sets for the PDDG and reoptimized NDDO methods, effort was made to considerably improve heats of formation without critical degradation of the other observables. MAEs for a set of 47 ionization potentials (Fig. 1) show that PDDG/PM3 yields the most accurate results, followed by PM3 and PM3'. Conversely, progression to MNDO' and PDDG/MNDO deteriorates the accuracy relative to MNDO for calculation of ionization potentials.

For a set of 44 dipole moments, MAEs of 0.34, 0.40, 0.29, and 0.35 D were obtained using PDDG/PM3, PDDG/MNDO, PM3', and MNDO', whereas errors of 0.32, 0.30, and 0.36 D were found

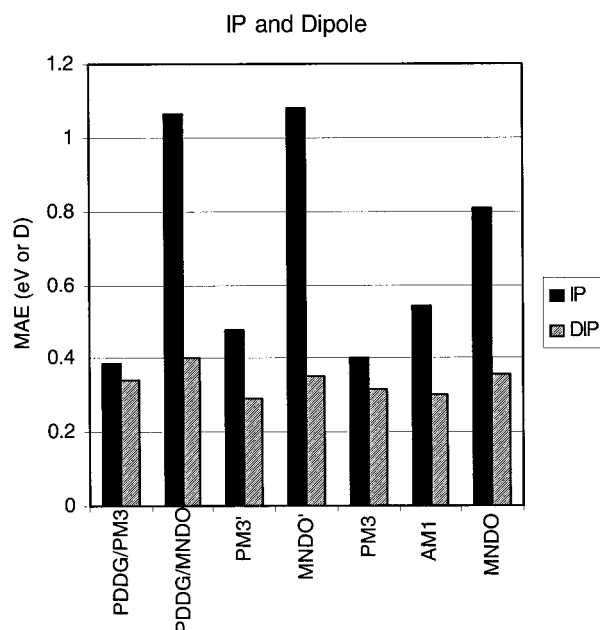
**Figure 1.** Mean absolute errors for ionization potentials (eV) and dipole moments (D) from the NDDO methods.

Table 11. Selected Ionization Potentials from PDDG and Reoptimized and Standard NDDO Methods (eV).

Molecule	PDDG		Reoptimized NDDO		Standard NDDO			Exp. ^a
	PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO	
Hydrogen fluoride	16.31	15.93	16.96	15.34	16.20	14.09	14.82	16.03
Fluoromethane	13.07	13.41	13.13	13.38	13.00	12.10	13.05	13.05
Fluoroethylene	10.74	10.54	10.69	10.30	10.65	10.24	10.17	10.58
Fluorine	16.14	16.66	16.42	16.05	15.61	14.28	15.25	15.83
<i>o</i> -Difluorobenzene	10.07	9.99	10.08	9.82	9.98	9.63	9.69	9.60
<i>m</i> -Difluorobenzene	10.12	10.04	10.13	9.86	10.02	9.72	9.73	9.60
<i>p</i> -Difluorobenzene	9.98	9.90	9.98	9.72	9.87	9.49	9.56	9.50
Oxygen fluoride	13.79	13.80	13.99	13.76	13.61	12.90	13.52	13.25
Carbonyl fluoride	13.59	13.47	13.48	13.75	13.38	13.39	13.61	13.62
Trifluoroethylene	10.86	10.96	10.86	10.68	10.94	10.09	10.46	10.53
Trifluoroacetic acid	12.40	12.50	12.32	12.64	12.25	12.25	12.50	11.77
Nitrogen trifluoride	12.26	14.71	12.33	14.46	12.24	13.18	13.93	13.73
Carbon tetrafluoride	16.99	17.65	17.52	17.06	16.81	15.32	16.81	16.25
Nitryl fluoride	13.98	12.64	13.66	13.29	13.59	13.39	12.99	13.51
Chlorine	10.95	12.86	10.91	12.82	10.72	11.59	12.45	11.59
Dichloromethane	10.80	12.71	10.94	12.85	10.62	11.39	12.49	11.40
Chlorine monoxide	10.79	12.64	10.80	12.54	10.39	11.86	12.29	11.02
Carbonyl chloride	11.65	12.72	11.62	12.99	11.20	12.37	12.70	11.84
Chloroform	11.25	13.25	11.35	13.37	10.81	11.77	12.91	11.48
Carbon tetrachloride	11.61	13.67	11.68	13.77	10.99	12.38	13.24	11.64
Tetrachloroethylene	9.46	10.96	9.66	10.93	9.50	9.90	10.73	9.51
Benzyl chloride	9.75	9.70	9.83	9.68	9.75	9.70	9.65	9.30
Chlorobenzene	9.45	9.76	9.53	9.67	9.42	9.56	9.62	9.31
Acetyl chloride	11.39	11.77	11.35	12.00	11.08	11.59	11.83	11.10
Hydrogen bromide	12.41	12.34	12.63	12.53	12.30	11.46	12.10	11.71
Bromomethane	11.00	11.69	11.10	11.92	11.02	10.80	11.56	10.54
Bromoethylene	10.41	10.54	10.45	10.45	10.39	10.15	10.25	9.80
Bromoethane	10.90	11.58	10.97	11.82	10.91	10.69	11.48	10.28
1-Bromopropane	10.91	11.57	10.98	11.80	10.93	10.69	11.47	10.18
Bromobenzene	9.75	9.73	9.81	9.67	9.71	9.60	9.55	9.25
Acetyl bromide	11.39	11.42	11.34	11.62	11.16	11.18	11.43	10.55
Cyanogen bromide	11.82	13.07	11.79	12.71	11.71	11.92	12.40	11.88
Bromine fluoride	11.64	13.08	11.67	13.18	11.64	11.77	12.65	11.87
Bromine chloride	11.00	12.40	10.97	12.48	10.92	11.22	12.01	11.10
Bromine	11.24	12.05	11.22	12.18	11.22	10.94	11.66	10.56
Dibromomethane	10.70	11.88	11.23	12.17	11.16	10.96	11.70	10.61
Carbonyl bromide	11.24	12.07	11.46	12.25	11.39	11.31	11.89	11.00
Bromoform	10.98	12.14	11.35	12.42	10.63	11.07	11.87	10.47
Benzyl bromide	9.66	9.62	9.73	9.64	9.73	9.60	9.56	9.23
Methyl iodide	9.63	11.12	9.94	11.18	9.48	10.51	10.85	9.54
Iodoethane	9.62	10.99	9.98	11.10	9.44	10.43	10.81	9.34
1-Iodopropane	9.62	10.99	9.97	11.08	9.45	10.43	10.80	9.27
Iodobenzene	9.18	9.80	9.33	9.71	9.12	9.65	9.55	8.78
Iodine chloride	9.78	12.21	9.69	12.24	9.74	11.12	11.58	10.10
Iodine bromide	9.97	11.89	10.06	11.97	9.84	10.84	11.28	9.85
Iodine	9.65	11.67	9.65	11.71	9.53	10.67	10.88	9.34
Diiodomethane	9.23	11.38	9.47	11.44	8.98	10.56	10.87	9.46

^aRef. 4.

for PM3, AM1, and MNDO (Fig. 1). Thus, there is little difference in quality for calculation of dipole moments among these methods. Tables 11 and 12 provide comparisons of selected ionization potentials and dipole moments given by PDDG and the reoptimized and standard NDDO methods with the experimental values.

Table 13 lists the MAEs in molecular geometries for all methods. Experimental and theoretical geometries were used as references, as mentioned above. MAEs for a set of 166 bond lengths and 44 bond angles reveal that PDDG/PM3 and PM3' yield small improvements over PM3 in the calculation of geo-

Table 12. Selected Dipole Moments from PDDG and Reoptimized and Standard NDDO Methods (Debye).

Molecule	PDDG		Reoptimized NDDO		Standard NDDO			Exp. ^a
	PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO	
Hydrogen fluoride	1.48	1.91	1.55	1.84	1.37	1.74	1.99	1.83
Fluoromethane	1.75	2.23	1.63	1.94	1.53	1.62	1.76	1.86
Fluoroacetylene	1.11	1.54	1.14	1.48	1.02	1.06	1.58	0.70
Fluoroethylene	1.62	1.99	1.56	1.84	1.39	1.38	1.70	1.43
Hypofluorous acid	1.73	2.11	1.86	1.78	1.73	1.60	1.81	2.23
1,1-Difluoroethane	2.42	2.95	2.28	2.70	2.12	2.27	2.50	2.30
<i>o</i> -Difluorobenzene	3.02	3.71	3.00	3.44	2.73	2.68	3.36	2.59
<i>m</i> -Difluorobenzene	1.76	2.15	1.73	1.99	1.57	1.55	1.93	1.58
Oxygen fluoride	0.43	0.54	0.69	0.39	0.39	0.11	0.32	0.30
Carbonyl fluoride	0.86	0.46	0.85	0.55	0.97	1.28	0.81	0.95
Trifluoroethylene	1.65	2.06	1.62	1.89	1.35	1.43	1.82	1.30
Trifluoroacetic acid	2.66	2.54	2.46	2.37	2.51	2.38	2.41	2.28
Nitrogen trifluoride	0.39	0.03	0.16	0.45	0.26	0.04	0.20	0.24
Trifluoromethyl hypofluorite	0.24	0.19	0.57	0.33	0.32	0.35	0.09	0.33
Cyanogen fluoride	1.82	1.43	1.61	0.97	1.74	1.21	0.89	2.17
Nitryl fluoride	0.20	0.08	0.14	0.19	0.25	0.82	0.66	0.47
Hydrogen chloride	1.71	1.68	1.82	1.63	1.38	1.38	1.48	1.12
<i>o</i> -Dichlorobenzene	1.78	2.95	2.12	3.09	1.34	1.97	2.84	2.26
Carbonyl chloride	1.32	0.54	0.96	0.28	0.93	1.28	0.47	1.17
1,1,1-Trichloroethane	1.78	2.29	1.89	2.31	1.35	1.75	2.08	1.76
Fluorotrichloromethane	0.57	0.58	0.37	0.29	0.56	0.20	0.11	0.46
Nitryl chloride	0.26	0.76	0.64	0.29	0.91	1.24	0.53	0.53
Chlorobenzene	1.21	1.95	1.41	2.02	1.00	1.31	1.86	1.72
Hydrogen bromide	1.58	1.36	1.51	1.28	1.24	1.38	1.07	0.83
Bromomethane	1.52	1.84	1.55	2.01	1.50	1.48	1.56	1.82
Bromoethylene	1.30	1.52	1.35	1.70	1.25	1.30	1.31	1.42
Bromoethane	1.77	1.99	1.84	2.14	1.74	1.66	1.66	2.03
1-Bromopropane	1.79	2.06	1.82	2.20	1.81	1.70	1.71	2.18
Bromobenzene	1.30	1.61	1.33	1.79	1.07	1.45	1.42	1.70
Acetyl bromide	2.71	2.62	2.76	2.89	2.94	2.59	2.45	2.43
Nitrosyl bromide	0.60	0.91	0.78	1.88	2.53	0.95	1.19	1.80
Bromine fluoride	2.05	2.01	2.52	1.81	2.25	1.46	2.10	1.42
Bromotrifluoromethane	1.24	1.79	1.15	1.27	0.91	1.02	1.18	0.65
Bromine chloride	0.39	0.87	0.53	0.83	0.06	0.45	0.75	0.52
Dibromomethane	1.57	1.69	1.38	1.78	1.23	1.32	1.37	1.43
Bromoform	1.15	1.17	0.93	1.21	1.00	0.91	0.91	0.99
Hydrogen iodide	1.68	1.28	0.78	1.02	0.83	1.27	1.02	0.44
Methyl iodide	1.56	1.75	1.62	1.91	1.83	1.35	1.37	1.65
Iodoethane	1.98	2.01	1.98	2.05	1.83	1.50	1.42	1.91
1-Iodopropane	1.97	2.06	1.94	2.11	1.78	1.54	1.47	2.04
Iodobenzene	1.12	1.69	1.16	1.76	1.06	1.43	1.60	1.70
Trifluoriodomethane	1.75	2.36	1.54	1.80	1.47	1.67	2.13	1.00
Iodine bromide	0.28	0.81	0.48	0.97	0.53	0.63	0.73	0.74
Diiodomethane	1.43	1.47	1.50	1.57	1.20	1.12	1.16	1.62

^aRef. 4.

metric parameters. While PDDG/MNDO and MNDO' generate more accurate bond lengths than AM1 and MNDO, the bond angles remain more accurate with the latter methods. MAEs over 41 bond lengths for complexes and eight bond lengths for the S_N2 transition structures, which are included in the overall MAEs in bond lengths, are also separately reported in Table 13. In this case, only theoretical results were used as references, including CCSD(T) and MP2 bond lengths from the literature,^{29,39,40} and MP2/6-311++G(d,p) and B3LYP/6-

311++G(d,p) bond lengths obtained in this work. The MAEs for both groupings show that the PDDG methods provide some improvement over the other NDDO methods in the calculation of molecular geometries for complexes and transition structures. In summary, the MAEs for ionization potentials, dipole moments, and molecular geometries show that the parameter sets for the PDDG and reoptimized NDDO methods do yield reductions in the MAEs for heats of formation without compromising the other important QM observables.

Table 13. Mean Absolute Errors for Molecular Geometries.

Geometric parameters	PDDG		Reoptimized NDDO		Standard NDDO			<i>N</i> ^a
	PM3	MNDO	PM3'	MNDO'	PM3	AM1	MNDO	
Overall								
Bond lengths (Å)	0.068	0.079	0.065	0.079	0.075	0.097	0.108	166
Bond angles (°)	3.1	3.8	2.7	3.9	3.7	2.2	3.4	44
Complexes								
Bond lengths (Å)	0.153	0.184	0.161	0.189	0.199	0.257	0.288	41
<i>S_N</i> 2 transition structures								
Bond lengths (Å)	0.080	0.130	0.098	0.165	0.124	0.217	0.210	8

^a*N* is the number of geometric parameters.

Conclusions

The PDDG/PM3 and PDDG/MNDO methods have been parameterized for fluorine, chlorine, bromine, and iodine. For comparison, the original MNDO and PM3 were also reoptimized for the halogens using the same training set, which provided the methods designated as MNDO' and PM3'. The PDDG and the reoptimized NDDO methods provide significant improvements over the standard semiempirical methods (PM3, AM1, and MNDO) with no increase in computation time. For a set of 442 molecules containing halogens, MAEs in heats of formation of 5.6, 6.6, 6.1, and 8.5 kcal/mol were found for PDDG/PM3, PDDG/MNDO, PM3', and MNDO'. This compares with 8.1, 11.1, and 14.0 kcal/mol for PM3, AM1, and MNDO. The errors in heats of formation with PDDG/PM3 and PM3' are not only smaller than those from prior NDDO methods, but in subsequent testing were also shown to outperform DFT with large basis sets for haloalkanes (Table 8). For the set of 155 normal-valent halogen-containing molecules selected by Thiel and Voityuk in their extension of MNDO to *d* orbitals,³⁵ MAEs in heats of formation of 3.9, 4.0, 4.1, 5.3, 5.3, and 6.3 kcal/mol were obtained for MNDO/*d*, PM3', MNDO', MNDO, AM1, and PM3, respectively. PDDG/PM3 and PDDG/MNDO, with MAEs of 3.3 and 3.4 kcal/mol, are the best performers for this data set. However, the development of the PDDG and reoptimized NDDO methods has not considered hypervalent halogen compounds and MNDO/*d* is the preferred procedure in this case.

Furthermore, PDDG/PM3 and PM3' give improved binding energies over the standard NDDO methods for complexes involving halide anions, and they are competitive with B3LYP/6-311++G(d,p) calculations including zero-point and thermal corrections (Table 9). Among the semiempirical methods studied, PDDG/PM3 generates the best agreement with *ab initio* G2 and CCSD(T) intrinsic activation energies for *S_N*2 reactions involving methyl halides and halide anions (Table 10). Finally, the MAEs in ionization potentials, dipole moments, and molecular geometries show that the PDDG and reoptimized NDDO methods yield much improved heats of formation without compromising the results for other important QM observables.⁴²

Supplementary Material

A table is provided listing the heats of formation from experiment and high-level *ab initio* calculations for the 442 halogen-containing molecules, together with the PDDG/PM3, PDDG/MNDO, PM3', MNDO', PM3, AM1, and MNDO results.

References

- Pople, J. A.; Beveridge, D. L. *Approximate Molecular Orbital Theory*; McGraw-Hill: New York, 1970.
- Dewar, M. J. S.; Thiel, W. *J Am Chem Soc* 1977, 99, 4899.
- Dewar, M. J. S.; Zoebisch, E. G.; Healy, E. F.; Stewart, J. J. P. *J Am Chem Soc* 1985, 107, 3902.
- Stewart, J. J. P. *J Comput Chem* 1989, 10, 221.
- Stewart, J. J. P. *J Comput Chem* 1989, 10, 209.
- Lyne, P. D.; Mulholland, A. J.; Richards, W. G. *J Am Chem Soc* 1995, 117, 11345.
- Ranganathan, S.; Gready, J. E. *J Phys Chem B* 1997, 101, 5614.
- Kaminski, G. A.; Jorgensen, W. L. *J Phys Chem B* 1998, 102, 1787.
- Antonczak, S.; Monard, G.; Ruiz-Lopez, M. F.; Rivail, J.-L. *J Am Chem Soc* 1998, 120, 8825.
- Proust-De Martin, F.; Dumas, R.; Field, M. J. *J Am Chem Soc* 2000, 122, 7688.
- Luque, F. J.; Reuter, N.; Cartier, A.; Ruiz-Lopez, M. F. *J Phys Chem A* 2000, 104, 10923.
- Reuter, N.; Dejaegere, A.; Maigret, B.; Karplus, M. *J Phys Chem A* 2000, 104, 1720.
- Marti, S.; Andres, J.; Moliner, V.; Silla, E.; Tunon, I.; Bertran, J.; Field, M. J. *J Am Chem Soc* 2001, 123, 1709.
- Gogonea, V.; Suarez, D.; van der Vaart, A.; Merz Jr., K. M. *Curr Opin Struc Biol* 2001, 11, 217.
- Faulder, P. F.; Tresaderm, G.; Chohan, K. K.; Scrutton, N. S.; Sutcliffe, M. J.; Hillier, I. H.; Burton, N. A. *J Am Chem Soc* 2001, 123, 8604.
- Chandrasekhar, J.; Shariffskul, S.; Jorgensen, W. L. *J Phys Chem B* 2002, 106, 8078.
- Repasky, M. R.; Guimaraes, C. R. W.; Chandrasekhar, J.; Tirado-Rives, J.; Jorgensen, W. L. *J Am Chem Soc* 2003, 125, 6663.
- Guimaraes, C. R. W.; Repasky, M. P.; Chandrasekhar, J.; Tirado-Rives, J.; Jorgensen, W. L. *J Am Chem Soc* 2003, 125, 6892.

19. Repasky, M. P.; Chandrasekhar, J.; Jorgensen, W. L. *J Comput Chem* 2002, 23, 1601.
20. Repasky, M. P.; Chandrasekhar, J.; Jorgensen, W. L. *J Comput Chem* 2002, 23, 498.
21. Tubert-Brohman, I.; Repasky, M. P.; Jorgensen, W. L. *J Comput Chem* 2003, submitted.
22. Stewart, J. J. P. *J Comput-Aided Mol Design* 1990, 4, 1.
23. Afeefy, H. Y.; Liebman, J. F.; Stein, S. E. In *NIST Chemistry Web-Book*; Linstrom, P. J.; Mallard, W. G., Eds.; National Institute of Standards and Technology: Gaithersburg, MD, 2001. <http://web-book.nist.gov/>.
24. Dewar, M. J. S.; Rzepa, H. S. *J Am Chem Soc* 1978, 100, 58.
25. Dewar, M. J. S.; Rzepa, H. S. *J Comput Chem* 1983, 4, 158.
26. Dewar, M. J. S.; Healy, E. *J Comput Chem* 1983, 4, 542.
27. Dewar, M. J. S.; Healy, E. F.; Stewart, J. J. P. *J Comput Chem* 1984, 5, 358.
28. Dewar, M. J. S.; Zoebisch, E. G. *J Mol Struct (Theochem)* 1988, 180, 1.
29. Glukhovtsev, M. N.; Pross, A.; Radom, L. *J Am Chem Soc* 1995, 117, 2024.
30. Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Zakrzewski, V. G.; Montgomery, J. A.; Stratmann, R. E.; Burant, J. C.; Dapprich, S.; Millam, J. M.; Daniels, A. D.; Kudin, K. N.; Strain, M. C.; Farkas, O.; Tomasi, J.; Barone, V.; Cossi, M.; Cammi, R.; Mennucci, B.; Pomelli, C.; Adamo, C.; Clifford, S.; Ochterski, J.; Petersson, G. A.; Ayala, P. Y.; Cui, Q.; Morokuma, K.; Malick, D. K.; Rabuck, A. D.; Raghavachari, K.; Foresman, J. B.; Cioslowski, J.; Ortiz, J. V.; Baboul, A. G.; Stefanov, B. B.; Liu, G.; Liashenko, A.; Piskorz, P.; Komaromi, I.; Gomperts, R.; Martin, R. L.; Fox, D. J.; Keith, T.; Al-Laham, M. A.; Peng, C. Y.; Nanayakkara, A.; Gonzalez, C.; Challacombe, M.; Gill, P. M. W.; Johnson, B.; Chen, W.; Wong, M. W.; Andres, J. L.; Gonzalez, C.; Head-Gordon, M.; Replogle, E. S.; Pople, J. A. *GAUSSIAN98*, Revision A.7; Gaussian, Inc.: Pittsburgh, PA, 1998.
31. Press, W. H.; Teukolsky, S. A.; Vetterling, W. T.; Flannery, B. P. *General Least Squares*, Press Syndicate of the University of Cambridge, New York, 1996, 1, 665.
32. Stewart, J. J. P. In *Encyclopedia of Computational Chemistry*; Schleyer, P. v. R.; Schreiner, P. R.; Schaefer III, H. F.; Jorgensen, W. L.; Thiel, W.; Glen, R. C.; Allinger, L. N.; Clark, T.; Gasteiger, J., Eds.; John Wiley & Sons: New York, 1998, pp 2000.
33. Stewart, J. J. P. In *Encyclopedia of Computational Chemistry*; Schleyer, P. v. R.; Schreiner, P. R.; Schaefer III, H. F.; Jorgensen, W. L.; Thiel, W.; Glen, R. C.; Allinger, L. N.; Clark, T.; Gasteiger, J., Eds.; John Wiley & Sons: New York, 1998, pp 2080.
34. Tikhomirov, V. A.; German, E. D.; Kuznetsov, A. M. *Chem Phys* 1995, 191, 25.
35. Thiel, W.; Voityuk, A. A. *Int J Quantum Chem* 1992, 44, 807.
36. Cloud III, C. F.; Schwartz, M. *J Comput Chem* 2003, 24, 640.
37. Larson, J. W.; McMahon, T. B. *J Am Chem Soc* 1983, 105, 2944.
38. Larson, J. W.; McMahon, T. B. *J Am Chem Soc* 1984, 106, 517.
39. Gonzales, J. M.; Cox III, R. S.; Brown, S. T.; Allen, W. D.; Schaefer III, H. F. *J Phys Chem* 2001, 105, 11327.
40. Schmatz, S.; Botschwina, P.; Stoll, H. *Int J Mass Spectrom* 2000, 201, 277.
41. Baushlicher, C. W. *Chem Phys Lett* 1995, 246, 40.
42. Directions for implementation of the PDDG methods in MOPAC are available at <http://zarbi.chem.yale.edu>. The PDDG methods are also available in the BOSS program; see www.cemcomco.com.